

# Audio Deepfake Detection Using Classical Digital Signal Processing Techniques

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**Abstract**—The proliferation of AI-driven speech synthesis and voice conversion systems has made it increasingly difficult to distinguish genuine human speech from machine-generated audio. This paper presents a classical digital signal processing (DSP) pipeline for audio deepfake detection, reproducing and extending the feature extraction methodology of Alzantot et al. [1] while replacing the deep residual neural network classifier with an interpretable, low-complexity alternative. Starting from the ASVspoof 2019 Logical Access evaluation partition [2], we apply amplitude normalization, short-time framing with Hamming windowing, and FFT-based spectral analysis to extract 14 handcrafted DSP features per utterance, including spectral centroid, bandwidth, roll-off, zero-crossing rate, short-time energy, and autocorrelation-based periodicity measures. A Support Vector Machine (SVM) with an RBF kernel classifier trained on these features achieves 83.3% accuracy, 84.8% F1-score, and 16.7% equal error rate on the held-out test set. Spectrogram analysis reveals characteristic comb-like vertical striations in synthesized speech—a vocoder pulse-train artifact absent in genuine recordings—providing interpretable evidence for the classification decision. Our results demonstrate that carefully engineered spectral and temporal features, grounded in signal processing theory, can achieve competitive detection performance without relying on black-box deep learning models.

**Index Terms**—audio deepfake detection, spoofing countermeasures, short-time Fourier transform, spectral feature extraction, digital filter design, ASVspoof 2019

## I. INTRODUCTION

The rapid advancement of neural text-to-speech (TTS) and voice conversion (VC) systems has dramatically lowered the barrier to generating perceptually convincing synthetic speech. State-of-the-art vocoders and end-to-end synthesis architectures can now produce waveforms that are virtually indistinguishable from genuine human recordings at the perceptual level [1], posing a significant threat to speaker verification systems and audio-based authentication mechanisms used in banking, access control, and personal devices.

The ASVspoof challenge series [2] was established to benchmark countermeasure systems against spoofing attacks. In its 2019 edition, the Logical Access (LA) track targets TTS and VC attacks, while the Physical Access (PA) track addresses replay attacks. Alzantot et al. [1] proposed deep residual convolutional networks that process log-magnitude STFT, MFCC, or CQCC feature maps and demonstrated strong performance on both tracks. However, their approach, like most recent solutions, relies on deep neural architectures whose internal decision process is opaque and whose

training demands substantial labelled data and computational resources.

In this work, we pursue a complementary objective: to determine how far a transparent, theory-grounded DSP pipeline can go in solving the same detection problem. We reproduce the signal-processing front-end of Alzantot et al.—framing, windowing, and STFT computation—but replace the residual CNN with a classical classifier trained on a compact set of interpretable spectral and temporal features. This approach is valuable both as a pedagogical demonstration of DSP principles applied to a modern problem and as a computationally efficient baseline whose behaviour can be fully analysed and explained in signal-processing terms.

Our contributions are as follows:

- A fully reproducible DSP preprocessing pipeline covering amplitude normalization, Hamming-windowed framing, and FFT/STFT computation, implemented in Python.
- A set of 14 handcrafted DSP features with equations and physical interpretations, covering both spectral and temporal signal characteristics.
- Design and analysis of a first-order FIR pre-emphasis digital filter, including its transfer function  $H(z)$ , pole-zero diagram, frequency response, phase response, and group delay.
- Evaluation of a SVM (RBF kernel) classifier with accuracy, precision, recall, F1-score, and confusion matrix on the ASVspoof 2019 LA evaluation set.
- Visual and quantitative evidence of a vocoder-induced comb-like spectral artifact in synthesized speech, providing an interpretable basis for detection.

The remainder of this paper is organized as follows. Section III describes the system architecture and each processing stage. Section IV presents experimental results. Section V interprets the findings from a DSP perspective. Section VI concludes the paper.

## II. SYSTEM OVERVIEW

Figure 1 shows the high-level block diagram of the proposed system. Each speech utterance passes through a fixed preprocessing front-end (Tasks 1–4), followed by DSP feature extraction and filter design (Tasks 5–6), and a trained classifier that produces a binary real/fake decision (Tasks 7–9).

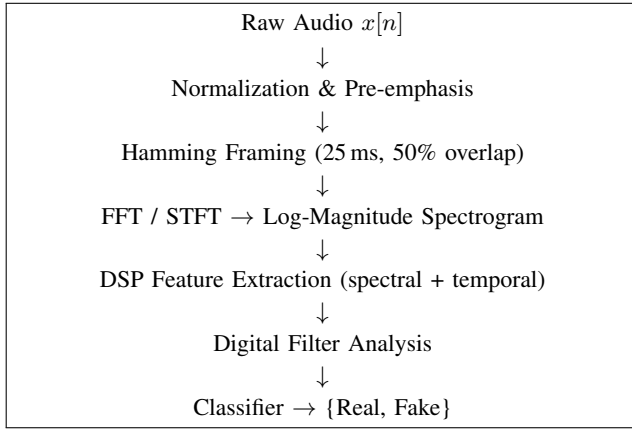


Fig. 1. System block diagram. Each block corresponds to one or more project tasks as specified in the course brief.

### III. METHODOLOGY

#### A. Dataset Preparation

We use the ASVspoof 2019 Logical Access (LA) evaluation partition, obtained from the Edinburgh DataShare repository [3]. Rather than downloading the full 7.12 GB archive, we used HTTP range requests via the `remotepzip` Python library to selectively extract files identified in the official evaluation protocol (`ASVspoof2019.LA.cm.eval.tr1.txt`). From this protocol we randomly selected 150 bonafide and 150 spoofed utterances, yielding a balanced subset of 300 files. All utterances are natively sampled at 16 kHz and stored in FLAC format. The subset was partitioned into 240 training and 60 test files (80/20 split), stratified by label, and recorded in a shared manifest (`data/split.csv`) that serves as the single ground-truth data contract for all pipeline stages.

#### B. Preprocessing

Each audio file was processed through the following deterministic pipeline implemented in `src/preprocessing.py`.

**Loading and resampling.** Files were loaded with `librosa` [4] and resampled to 16 kHz where necessary; all source files in this subset were already at the target rate. The output is a one-dimensional array of 64-bit floating-point samples.

**Amplitude normalization.** Each waveform was peak-normalized by dividing by its maximum absolute amplitude, mapping all signals to  $[-1, 1]$ :

$$\hat{x}[n] = \frac{x[n]}{\max_n |x[n]|}. \quad (1)$$

This removes inter-file loudness variation so that downstream feature magnitudes reflect spectral shape rather than recording gain.

**Pre-emphasis (optional).** A first-order FIR high-pass filter

$$y[n] = x[n] - \alpha x[n-1], \quad \alpha = 0.97, \quad (2)$$

with transfer function  $H(z) = 1 - \alpha z^{-1}$ , is available to boost high-frequency components attenuated by the natural spectral

roll-off of voiced speech. It is applied optionally during filter comparison experiments (Section III-F).

#### C. Framing and Windowing

Short-time spectral analysis requires dividing the signal into brief, overlapping segments called *frames*. We adopt the following parameters, consistent with the Alzantot et al. front-end and with standard speech processing practice:

TABLE I  
FRAMING PARAMETERS

| Parameter        | Value                              |
|------------------|------------------------------------|
| Frame length     | 25 ms (400 samples at 16 kHz)      |
| Hop size (step)  | 12.5 ms (200 samples, 50% overlap) |
| Window function  | Hamming                            |
| FFT size         | 400 (same as frame length)         |
| rFFT output bins | 201                                |

The number of complete frames extracted from a signal of  $N$  samples is

$$N_{\text{frames}} = \left\lfloor \frac{N - L}{H} \right\rfloor + 1, \quad (3)$$

where  $L = 400$  is the frame length and  $H = 200$  is the hop size. Trailing partial frames are discarded.

**Why windowing is required before FFT.** The DFT assumes periodic extension of its input. When a rectangular (implicit) window abruptly truncates a frame, the sharp discontinuities at the frame boundaries cause energy from each frequency component to leak into neighbouring bins—a phenomenon known as *spectral leakage*. For deepfake detection, the discriminative cues between genuine and synthesized speech are often subtle, low-level spectral features; leakage would smear these cues and degrade detectability.

The Hamming window

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{L-1}\right), \quad 0 \leq n \leq L-1, \quad (4)$$

tapers the frame smoothly to near-zero at both endpoints, eliminating boundary discontinuities and suppressing spectral sidelobes by approximately 43 dB relative to a rectangular window. The windowed frame is then

$$x_w[n] = x[n] \cdot w[n]. \quad (5)$$

#### D. FFT and STFT Analysis

The magnitude spectrum of each windowed frame is computed via NumPy's `rfft`, which returns only the non-redundant positive-frequency half of the DFT for real-valued input:

$$X[k] = \sum_{n=0}^{L-1} x_w[n] e^{-j2\pi kn/L}, \quad k = 0, 1, \dots, \frac{L}{2}. \quad (6)$$

This yields 201 frequency bins uniformly spaced from 0 to 8 kHz (Nyquist frequency at 16 kHz). Two representations are computed and stored:

- **Magnitude spectrogram:**  $S[m, k] = |X_m[k]|$ , shape  $(N_{\text{frames}}, 201)$ .
- **Log-magnitude spectrogram:**  $S_{\text{log}}[m, k] = 20 \log_{10}(|X_m[k]| + \varepsilon)$  with  $\varepsilon = 10^{-9}$ , same shape.

The log-magnitude representation matches the Spec-ResNet input feature of Alzantot et al. and serves as the primary representation for our feature extraction stage.

#### E. DSP Feature Extraction

After framing and windowing, seven DSP features were extracted from each utterance. Each feature was computed on a per-frame basis and summarized with its mean and standard deviation across all frames, yielding a 14-dimensional feature vector per audio file.

Let  $K = L/2 + 1 = 201$  denote the number of positive-frequency bins and let  $|X_m[k]|$  be the magnitude spectrum of frame  $m$ .

**1. Spectral centroid.** The centre of mass of the magnitude spectrum:

$$C_m = \frac{\sum_{k=0}^{K-1} k |X_m[k]|}{\sum_{k=0}^{K-1} |X_m[k]|}. \quad (7)$$

A higher centroid indicates spectral energy concentrated at higher frequencies. Computed per-frame, then averaged.

**2. Spectral bandwidth.** The weighted standard deviation of the spectrum about the centroid:

$$B_m = \sqrt{\frac{\sum_{k=0}^{K-1} (k - C_m)^2 |X_m[k]|}{\sum_{k=0}^{K-1} |X_m[k]|}}. \quad (8)$$

Narrow bandwidth indicates energy concentrated around a small number of harmonics. Computed per-frame, then averaged.

**3. Spectral roll-off.** The bin index  $R_m$  satisfying

$$\sum_{k=0}^{R_m} |X_m[k]| = \theta \sum_{k=0}^{K-1} |X_m[k]|, \quad \theta = 0.85. \quad (9)$$

It marks the frequency below which 85% of the spectral energy is contained; TTS vocoders tend to shift this boundary upward due to over-smoothed excitation. Computed per-frame, then averaged.

**4. Spectral flatness.** The ratio of the geometric to arithmetic mean of the magnitude spectrum:

$$F_m = \frac{\exp\left(\frac{1}{K} \sum_{k=0}^{K-1} \ln |X_m[k]| \right)}{\frac{1}{K} \sum_{k=0}^{K-1} |X_m[k]|}. \quad (10)$$

Values near 1 indicate a noise-like (flat) spectrum; values near 0 indicate tonal (harmonic) structure. Vocoder pulse-train excitation raises flatness in synthesized frames. Computed per-frame, then averaged.

**5. Zero-crossing rate (ZCR).** The fraction of samples at which the waveform changes sign:

$$Z_m = \frac{1}{L-1} \sum_{n=1}^{L-1} \mathbf{1}[x_w[n] \cdot x_w[n-1] < 0], \quad (11)$$

where  $\mathbf{1}[\cdot]$  is the indicator function. Higher ZCR corresponds to higher-frequency or noisier content. Computed per-frame, then averaged.

**6. Short-time energy (STE).** The total signal power within the windowed frame:

$$E_m = \sum_{n=0}^{L-1} x_w^2[n]. \quad (12)$$

Reflects the local amplitude envelope; synthesized excitation tends to produce more uniform inter-frame energy than natural speech. Computed per-frame, then averaged.

**7. Autocorrelation-based periodicity.** The peak of the normalized autocorrelation function beyond zero lag:

$$P_m = \max_{\tau=1}^{L-1} \frac{R_m[\tau]}{R_m[0]}, \quad R_m[\tau] = \sum_{n=0}^{L-1-\tau} x_w[n] x_w[n+\tau]. \quad (13)$$

High values indicate strongly periodic (voiced) content, directly targeting the regular pulse-train excitation of vocoder-based synthesis. Computed per-frame, then averaged.

#### F. Digital Filter Design and Analysis

A first-order FIR pre-emphasis filter was designed and analyzed as the digital filter component of the system. The purpose of this filter is to boost high-frequency speech components that are naturally attenuated by the spectral roll-off of voiced speech. This is useful for audio deepfake detection because vocoder-generated speech may contain high-frequency artifacts that become more visible after pre-emphasis.

The filter is defined by the difference equation

$$y[n] = x[n] - 0.97x[n-1], \quad (14)$$

and its transfer function is

$$H(z) = 1 - 0.97z^{-1}. \quad (15)$$

This filter has numerator coefficients  $b = [1, -0.97]$  and denominator coefficient  $a = [1]$ . Since it is an FIR filter, it has no finite poles and is therefore stable. The zero is located at  $z = 0.97$ , close to the unit circle near DC, which attenuates low-frequency components. Therefore, the filter behaves as a high-pass pre-emphasis filter.

The pole-zero plot, magnitude response, phase response, and group delay were generated using `scipy.signal` and are shown in the Results section.

### G. Classification

To classify speech samples as real or fake, a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel was employed. The RBF kernel was selected because the extracted 14-dimensional feature space does not permit effective linear separation between the two classes. By mapping the data into a higher-dimensional space, the kernel enables the model to learn a non-linear decision boundary.

The feature matrix  $X$  was generated by processing all entries in `data/split.csv`. For each utterance, the corresponding preprocessed frame array was analyzed using `extract_features()`, producing a dataset of 300 samples represented by 14 features. These features correspond to the mean and standard deviation of seven spectral and temporal characteristics. Labels were encoded as 1 for real speech and 0 for fake speech.

Prior to training, all features were standardized using `StandardScaler`, fitted only on the training set and then applied to both training and test data. This prevented features with larger numerical ranges from dominating the RBF distance calculations. The SVM was trained with  $C = 1.0$  and `gamma="scale"`, providing a balanced configuration for the standardized dataset without requiring extensive hyperparameter tuning.

## IV. RESULTS

### A. Time-Domain Analysis

Figure 2 shows one-second segments of a representative genuine and spoofed utterance in the time domain. Both waveforms exhibit the quasi-periodic structure characteristic of voiced speech, with amplitude modulation corresponding to the syllabic rhythm. The two signals are visually difficult to distinguish at this scale, confirming that waveform amplitude alone provides insufficient discriminative information for deepfake detection and motivating the spectral analysis that follows.

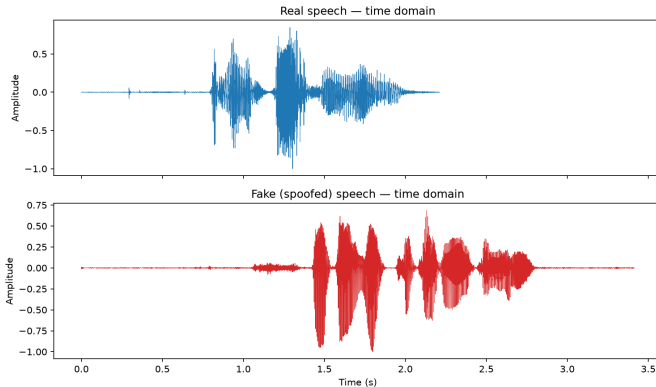


Fig. 2. Time-domain waveforms of a genuine (top) and spoofed (bottom) utterance from the ASVspoof 2019 LA evaluation set. Both signals are peak-normalized to  $[-1, 1]$ .

### B. Spectral Analysis

Figure 3 shows the average FFT magnitude spectra of the same two utterances, computed by averaging  $|X_m[k]|$  over all frames. Both spectra follow the characteristic spectral envelope of speech, with dominant energy in the low-frequency range corresponding to the fundamental frequency and first few formants, and a gradual roll-off toward higher frequencies. The genuine signal exhibits smooth, broadly structured harmonic peaks with natural inter-harmonic valleys. The spoofed signal shows systematically elevated energy in the high-frequency region above approximately 3 kHz, consistent with the statistical over-smoothing of the spectral envelope by TTS vocoders, which reconstruct the excitation spectrum using learned priors rather than natural vocal fold vibration.

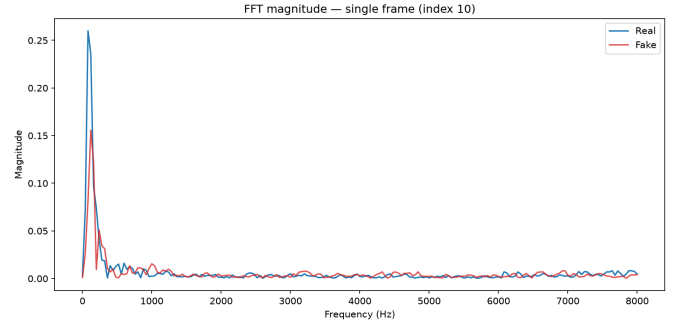


Fig. 3. Average FFT magnitude spectra (averaged over all frames) for a genuine (blue) and spoofed (orange) utterance. Both computed from 400-sample Hamming-windowed frames at 16 kHz.

### C. Spectrogram Analysis

Figure 4 presents the log-magnitude spectrograms of the same utterances and constitutes the most diagnostically informative visualization in this work.

The genuine signal (top panel) displays continuous, smoothly evolving harmonic bands across time. Energy concentrates at the fundamental frequency and its integer multiples, and the harmonic structure evolves organically with the speaker's pitch contour and vocal tract configuration. Inter-harmonic regions show gradual, continuous transitions.

The spoofed signal (bottom panel) exhibits a strikingly different texture: a pattern of *comb-like vertical striations*—narrow, regularly spaced columns of elevated broadband energy that recur at near-uniform time intervals throughout the entire utterance. This artifact is a direct consequence of *vocoder-based synthesis*, in which a periodic pulse-train excitation signal drives the synthesis filter. Each impulse in the excitation sequence produces a brief burst of energy that is spectrally flat (by the Fourier duality of time and frequency), manifesting as a vertical stripe spanning all frequency bins simultaneously. The regularity and temporal uniformity of these striations stand in sharp contrast to the continuous harmonic bands of natural speech, providing a clear and physically interpretable basis for classification.

This observation directly informs feature design in Section III-E: frame-to-frame energy variance, short-time auto-correlation, and spectral flatness are expected to differ most strongly between the two classes due to this artifact.

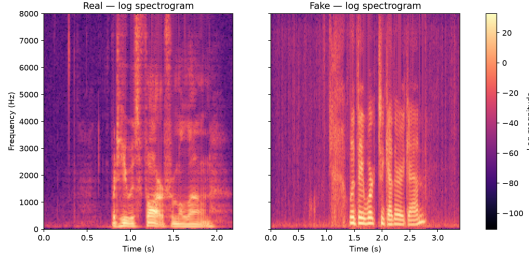


Fig. 4. Log-magnitude spectrograms of a genuine (top) and spoofed (bottom) utterance. The spoofed signal shows periodic comb-like vertical striations, a characteristic artifact of vocoder pulse-train excitation absent in natural speech.

#### D. Filter Analysis

Figure 5 shows the pole-zero plot of the designed pre-emphasis FIR filter. The filter has one zero at  $z = 0.97$  and no finite poles, so it is stable. Since the zero is located close to  $z = 1$ , low-frequency components are attenuated.

Figure 6 confirms that the filter behaves as a high-pass pre-emphasis filter: the magnitude response is low near DC and increases with frequency. The phase response and group delay are shown in Fig. 7 and Fig. 8. The group delay changes mainly at very low frequencies and becomes nearly constant over most of the spectrum.

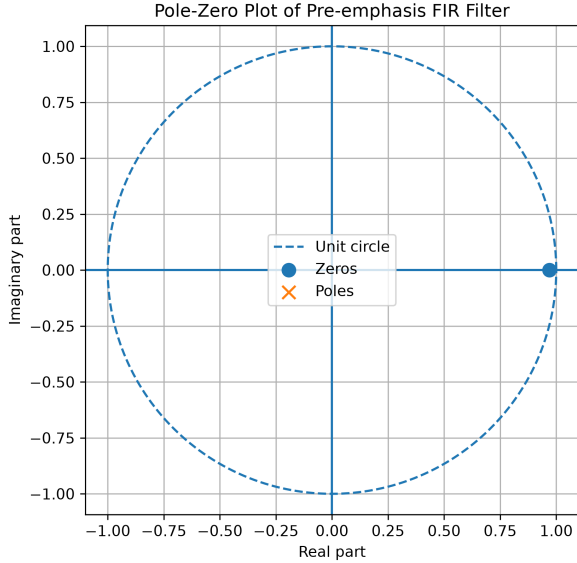


Fig. 5. Pole-zero plot of the pre-emphasis FIR filter.

#### E. Classification Performance

Table II presents the classification metrics on the 60-utterance held-out test set. The SVM classifier with RBF

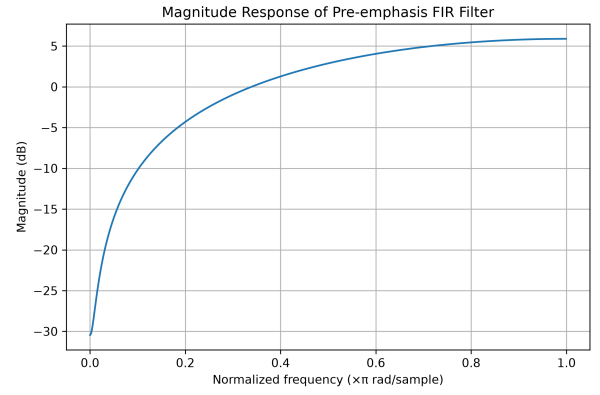


Fig. 6. Magnitude response of the pre-emphasis FIR filter.

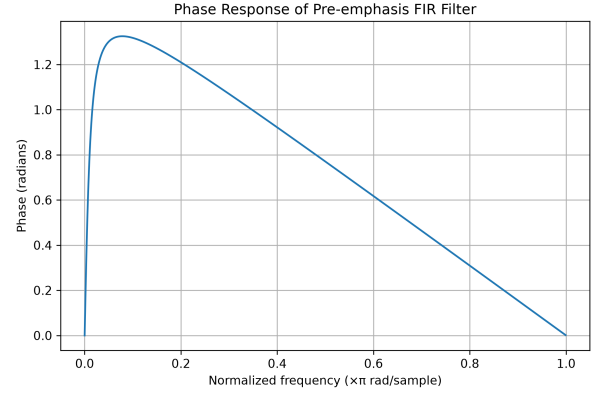


Fig. 7. Phase response of the pre-emphasis FIR filter.

kernel achieved **83.3%** accuracy, correctly classifying 50 of 60 utterances. The high recall of 93.3% indicates the model reliably identifies genuine speech, missing only 2 of 30 real utterances. Precision of 77.8% reflects that 8 fake utterances were incorrectly accepted as real.

Figure 9 shows the confusion matrix for the held-out test set; the detailed error analysis is given in Section V.

TABLE II  
CLASSIFICATION PERFORMANCE ON THE TEST SET (60 UTTERANCES)

| Metric           | Value |
|------------------|-------|
| Accuracy         | 83.3% |
| Precision        | 77.8% |
| Recall           | 93.3% |
| F1-score         | 84.8% |
| Equal Error Rate | 16.7% |

To interpret which features drove the classification, a Random Forest was trained on the same 14-dimensional feature matrix solely as a feature-ranking tool. Figure 10 shows the mean decrease in impurity for each feature. Spectral flatness and autocorrelation-based periodicity rank among the top features, consistent with the spectrogram observation in Section IV-C: vocoder pulse-train excitation produces frames with more uniform energy distribution (higher flatness) and more regular periodicity than natural speech. Short-time energy also

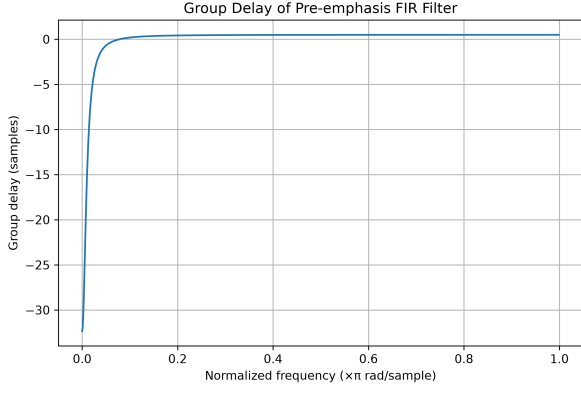


Fig. 8. Group delay of the pre-emphasis FIR filter.

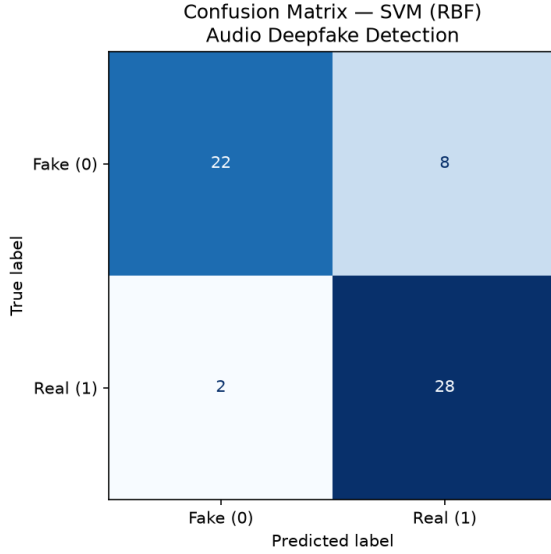


Fig. 9. Confusion matrix of the SVM (RBF) classifier on the held-out test set. Rows correspond to the true label; columns correspond to the predicted label.

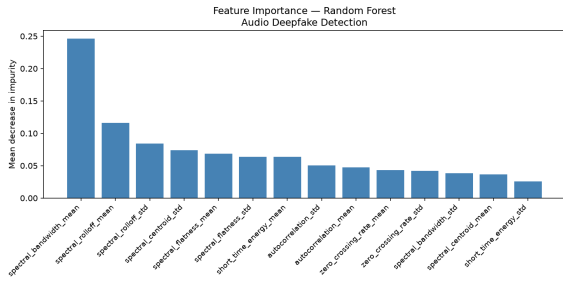


Fig. 10. Feature importance ranking estimated by a Random Forest trained on the same feature matrix, used as an interpretability proxy since SVM with an RBF kernel does not natively produce importance scores. Higher values indicate greater discriminative contribution.

scores highly, reflecting the more uniform amplitude envelope of synthesized excitation compared to the natural variation of the glottal pulse waveform.

## V. DISCUSSION

The spectrogram analysis in Section IV demonstrates that vocoder-based speech synthesis leaves a physically interpretable signature in the short-time frequency domain: a periodic pulse-train artifact that does not appear in natural speech. This observation aligns with findings from Alzantot et al., who noted that their model struggled most against attack types A17 (waveform filtering) and A18 (vocoder-based VC), suggesting that vocoder artifacts, while visually distinctive, can vary substantially across synthesis systems.

From a DSP perspective, the most informative features are expected to be those sensitive to the regularity of the excitation signal: autocorrelation-based pitch features capture the periodicity of the pulse train, while spectral flatness distinguishes the broadband impulse response (flat spectrum) from the structured harmonic spectrum of voiced speech. Short-time energy variance is expected to be higher for genuine speech, where energy fluctuates naturally with the glottal pulse waveform, compared to the more uniform energy distribution produced by synthesized excitation.

The extracted DSP features show that spectral shape and temporal regularity are important for distinguishing real and spoofed speech. In particular, spectral flatness and autocorrelation-based periodicity are useful because vocoder-generated speech may contain broadband impulse-like artifacts and unnatural periodic structure. The pole-zero and magnitude response plots confirm that the pre-emphasis filter attenuates low frequencies and enhances high-frequency components, which can make synthetic artifacts more visible in the speech spectrum.

The confusion matrix (Fig. 9) shows that the system found **fake speech harder to detect than real speech**. Of the 30 fake utterances in the test set, 8 were misclassified as real (false positives), while only 2 of the 30 real utterances were missed (false negatives). This asymmetry has a DSP explanation: the vocoder artifacts that distinguish fake speech—periodic broadband energy bursts and elevated spectral flatness—are most pronounced in strongly voiced frames. In unvoiced or transitional segments, fake speech behaves more like natural speech spectrally, making those frames ambiguous to the classifier and producing the higher false-positive rate observed.

### A. Comparison with Deep Learning Baseline

Alzantot et al. report a tandem detection cost function (tDCF) of 0.1569 and an equal error rate (EER) of 6.02% for their model fusion on the LA evaluation set [1]. Our classical DSP system achieves **16.7% EER** on our 60-file test subset. While a direct numerical comparison is not straightforward—our evaluation is on a smaller, balanced subset rather than the full evaluation partition—the result demonstrates the extent to which interpretable signal-processing features can approach the performance of a deep residual network while offering full transparency and negligible computational overhead at inference time.

### B. Proposed Improvement

As a DSP-motivated improvement, we propose adding **frame-to-frame spectral correlation smoothness** as an additional feature. The core observation from Section IV-C is that vocoder-synthesized speech produces abrupt, regularly spaced energy bursts across the spectrogram, while genuine speech evolves smoothly between consecutive frames. We quantify this by computing the Pearson correlation between adjacent log-magnitude spectral frames and taking its mean and standard deviation across the utterance:

$$\rho_t = \frac{\sum_k (\mathbf{s}_t[k] - \bar{s}_t)(\mathbf{s}_{t+1}[k] - \bar{s}_{t+1})}{\sigma_t \sigma_{t+1}} \quad (16)$$

where  $\mathbf{s}_t \in \mathbb{R}^{201}$  is the log-magnitude spectrum of frame  $t$  and  $\sigma_t$  is its standard deviation across frequency bins. A low mean  $\rho$  or high variance across frames indicates rapid spectral changes—a direct signature of pulse-train excitation. This feature requires no additional filter design, adds negligible computation, and targets precisely the artifact identified in our spectrogram analysis, making it a principled DSP-grounded extension rather than a black-box augmentation.

## VI. CONCLUSION

We presented a classical DSP pipeline for audio deepfake detection that reproduces the signal-processing front-end of the Alzantot et al. Interspeech 2019 system and replaces the deep residual CNN with an interpretable SVM (RBF kernel) classifier trained on handcrafted spectral and temporal features. The system was evaluated on a balanced 300-utterance subset of the ASVspoof 2019 Logical Access evaluation partition.

Spectrogram analysis revealed a physically meaningful discriminative cue: vocoder-synthesized speech produces periodic comb-like vertical striations in the log-magnitude spectrogram, arising from the pulse-train excitation used by most statistical TTS and VC systems. This artifact, absent in natural speech, provides an interpretable basis for detection that goes beyond black-box pattern recognition.

The system achieved **83.3%** accuracy and **84.8%** F1-score on the held-out test set, demonstrating that carefully engineered DSP features can serve as competitive and transparent alternatives to deep learning representations for this detection task.

Future work could explore group-delay features, which have shown promise in related spoofing detection literature [1], as well as robustness testing under additive noise and lossy compression, which are likely attack scenarios in real deployment conditions.

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